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# PHOTOGRAPHIC PHOTOMETRY WITH IRIS DIAPHRAGM PHOTOMETERS

BRADLEY E. SCHAEFER*
Center for Space Research, MIT Room 37-576, Massachusetts Institute of Technology,
Cambridge, Massachusetts 02139

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This paper provides a general method for solving problems which commonly arise while using an iris diaphragm photometer (IDP). One application of this method is a derivation of the general shape of the calibration curve. Another application is a procedure for correcting for the effects of variable background fog level across the plate. The method consists of (1) assuming the profile of starlight incident on the plate, (2) assuming the shape of the plate's characteristic curve, (3) using (1) and (2) to derive the profile of the star image, and (4) determining what the IDP reading will be for each star image. The result is a relation between the IDP reading, the magnitude of the star, and three parameters of the plate-telescope system. These parameters can be evaluated in many ways, for example, by applying the relation to comparison stars. This method requires only an IDP, a sequence of comparison stars on the plate, and easily performed calculations. The method is also used to determine a relation between the radius of a photographic star image and the star's magnitude, and to predict the IDP reading of the "point of optimum density." The results of this article are illustrated using several data sets taken from the literature.

*Key words:* photographic photometry—iris diaphragm photometers—calibration curve—point of optimum density method

## I. Introduction

Photographic photometry has many uses and large amounts of data have been measured with iris diaphragm photometers (IDP). This paper develops a general method for solving various problems encountered in the analysis of IDP data. The most useful application of this method is to find the general form of the calibration curve (the relation between iris radius and stellar magnitude). Another application is in developing a procedure for correcting for a variable background fog level.

It is desirable to know the form of the calibration curve so that data may be better fit or so that the curve may be extrapolated with greater confidence. Several authors have tried to find this functional dependence. Argue (1960) and Burkhead and Seeds (1971) have fitted polynomials up to twelfth order to the calibration curve. Stetson and Harris (1977) have fitted series of Chebyshev polynomials plus a hyperbolic term to the calibration curve. This approach is good for interpolation, but has no physical motivation and cannot be used for extrapolation. Stock (1958) has presented a relationship for bright stars (eq. (17) of this paper). His result applies only to the IDP type which uses a variable iris size (hereafter called a VIDP), but not to an IDP type which uses a fixed iris size (hereafter called a FIDP). Moffat (1969) also derived a calibration curve for the VIDP, but his complex solution has six parameters (his sixth parameter is the equivalent of $R_B$) several of which are to be determined with a microdensitometer. Brown (1978) used a computer to numerically find the calibration curve for stars and faint galaxies. He modeled the profile of incoming starlight as the sum of three Gaussians (cf. section II, part A of this paper), and used microdensitometry to evaluate model parameters.

Ross (1936) has derived an easy method to correct for the effects of variable background fog on FIDP measurements. Dixon (1970) has examined Ross's method and finds it satisfactory. Chun (1976) has shown Ross's method to be correct for a calibration curve like that given in equation (5) of this paper. Argue (1960), Purbosiswojo (1963), and Schaefer (1979) all discuss empirical methods for correcting for variable background in VIDP measurements. Weaver (1962) gives a formula for correcting for background variations for a VIDP with large iris openings. Weaver (1947) gives an empirical approach to the "point of optimum density" method of photographic transfer of sequences of comparison stars.

Section II of this paper outlines the general method to be used for relating a star's magnitude to its IDP reading. This procedure is carried out in section three, where the general form of the calibration curve is found. The results of this section are summarized in Table III. Section IV demonstrates a procedure for correcting for the effects of a nonconstant background fog level on the plate. The appendix gives the solutions to several problems encountered while using an IDP. For example, formulae are given relating the radius of a star's image to its magnitude, and for finding the IDP reading of the "point of optimum density." An effort has been made to keep the results practical enough for general use: (1) Only an IDP is needed. (2) No special plate material is needed (for example, sensitometric spots are not re-

*Work begun while with Department of Astronomy, University of California at Berkeley.

quired). However, the plate must have a sequence of comparison stars of known magnitude. (3) The mathematics is simple enough to be quickly soluble on a programmable pocket calculator.

The casual reader, who is more concerned with using the predictions of this method, is advised to read the last two paragraphs of section III and refer to Table III.

## II. The Method

The method presented in this paper consists of assuming a profile for a stellar image incident on the plate and assuming the shape of the plate's characteristic curve. The transmittance of the stellar image on the plate can then be found, and this is used (along with the knowledge of how an IDP works) to calculate the IDP reading for that star. The modeled IDP reading will be a function of stellar magnitude and various parameters of the plate-telescope system. The comparison stars are used to evaluate these parameters. Then the equation can be applied to the star of interest to find the star's magnitude (which is usually the desired quantity) as a function of the measured iris radius.

The purpose of this paper is to develop a simple model of calibration curves which yields accurate results in an analytic form. There are many effects that will cause small changes in the shape of the calibration curve. Some examples of these small effects are the effects due to coma, finite grain size, the shoulder of the characteristic curve, and spikes on the star image. In keeping with the purpose of this paper, I have not attempted to model these effects because they are insignificant and difficult to model.

A. *Stellar Profile.* In most practical cases, the total energy acting on a unit area of the emulsion ($I$) is closely approximated by a two-dimensional Gaussian (Gyldenkaerne 1950) due to random scattering of photons by the atmosphere.

$$\boxed{I(r,\ell) = \frac{\ell}{2\pi R_*^2} \exp\left(\frac{-r^2}{2R_*^2}\right)} \qquad (1)$$

(all variables and parameters are defined in Table I). But far out in the wings, other effects are dominant. King (1971) gives a complete star profile, which is adopted for use in this paper. Outside of the Gaussian region, an exponential profile fits well,

$$I(r,\ell) = \beta\ell e^{-r/\sigma} \quad . \qquad (2)$$

Still farther out, the profile is well fit by various power laws,

$$I(r,\ell) = \frac{\delta\ell}{r^\eta} \quad . \qquad (3)$$

De Vaucouleurs (1958), Van de Hulst (1952), King (1971), and Kormendy (1973) find values for $\eta$ ranging between 2 and 5, and discuss possible causes for this region of the profile.

B. *Characteristic Curve.* The developed plate has a local transmission ($T$) which is determined by the exposure ($I$, in units of energy per area). The function of $T$ versus $I$ is called the characteristic curve of the plate. (Traditionally, the characteristic curve is a plot of $\log I$ versus density, where density equals $-\log T$, but $T$ and $I$ are used in this paper as they are more convenient.) The characteristic curve is complicated and must be modeled. The simplest model is to assume that the curve is linear;

$$T(I) = \begin{cases} 0 & ; \quad 1 - \lambda I < 0 \\ 1 - \lambda I & ; \quad \text{otherwise} \\ T_M & ; \quad 1 - \lambda I > T_M \quad . \end{cases} \qquad (4)$$

A better model is that $T$ and $I$ are related by a power law;

$$T(I) = \begin{cases} T_M & ; \quad (\alpha I)^{-\gamma} > T_M \\ (\alpha I)^{-\gamma} & ; \quad (\alpha I)^{-\gamma} < T_M \quad . \end{cases} \qquad (5)$$

Models of characteristic curves can be made quite complex (see for example Stock 1958; Tsubaki and Engvold 1975), yet even the simple linear model is sufficiently accurate for most practical cases.

C. *Iris Diaphragm Photometer.* Two types of IDP's are used. For the FIDP (Weaver 1962; Ross 1936), the amount of light passing through the plate and a fixed iris is measured by a photomultiplier:

$$P = \chi \int_0^{2\pi} \int_0^{R_R} T(r)\, r\, dr\, d\theta \quad . \qquad (6)$$

If there is not a star in the area of the plate measured ($T(r) = T_B$), then a reading of the background is made:

$$P_B = \chi \pi R_R^2 T_B \quad . \qquad (7)$$

Ross (1936) showed that the quantity

$$Q = P/P_B \qquad (8)$$

is a background independent measure of a star's brightness.

For a VIDP (Weaver 1962; Argue 1960; Cameron 1951), the size of the iris is varied until the intensity of light passing through the plate and iris equals the intensity of a comparison beam:

$$\kappa = \int_0^{2\pi} \int_0^{R_s} T(r)\, r\, dr\, d\theta \quad . \qquad (9)$$

$R_s$ is the measure of the star's brightness. If there is no star in the field of view of the iris ($T(r) = T_B$), then

$$\kappa = \pi R_B^2 T_B \quad . \qquad (10)$$

$\kappa$ may be determined by making a reading with no plate

TABLE I

Definition of Variables

| | | |
|---|---|---|
| $r,\theta,x,y$ | (1,38) | polar and cartesian coordinates (origins at star's center) |
| $a$ | (38) | ratio of minor to major axis of trailed star image |
| $I$ | (1–3) | total quantity of energy acting on the emulsion per unit area ("exposure") |
| $\ell$ | (1–3,15) | total quantity of energy from a star acting on the emulsion |
| $\ell_J$ | (15) | $I$ for a star of magnitude $m(\ell_J)$ |
| $m$ | (15) | magnitude of star |
| $m(\ell_J)$ | (15) | $m$ for star whose image is just barely saturated in the center, and which is in the Gaussian regime |
| $P$ | (6) | photomultiplier reading from a FIDP |
| $Q$ | (8) | background-independent measure of $m$ for a FIDP |
| $Q_0$ | | value of $Q$ at the "point of optimum density" |
| $R_*$ | (1) | Gaussian radius of starlight striking plate |
| $R_{1/2}$ | (36) | radius of unsaturated stars profile at half maximum |
| $R_2$ | (29) | $R_s$ when more than one star in iris' field of view |
| $R_B$ | (10) | VIDP reading when no star is in the iris' image |
| $R_C$ | (29) | $R_s$ for a "nearby star" of known $m$ |
| $R_I$ | (31) | radius at which a star image's profile has $T$ = const |
| $R_0$ | (40) | $R_s$ value at "point of optimum density" |
| $R_R$ | (6) | fixed iris radius for a FIDP |
| $R_s$ | (9) | VIDP reading when a star is centered in iris' image |
| $R_T$ | (39) | $R_s$ value for a trailed star image |
| $R_{\mathrm{TOE}}$ | | $r$ at which star's image becomes indistinguishable from background |
| $T$ | (4,5) | transmission coefficient of the developed plate |
| $T_M$ | (4,5) | maximum $T$ which the plate can have |
| $W$ | (12) | variable which is proportional to $\ell$ |
| $\alpha$ | (5) | parameter of plate sensitivity |
| $\beta$ | (2) | parameter of star profile |
| $\gamma$ | (5) | gamma, slope of the $-\log T$ vs. $\log I$ curve (H.D. curve) |
| $\delta$ | (3) | parameter of star's profile |
| $\eta$ | (3) | power by which star profile varies with $r^{-1}$ |
| $\kappa$ | (9) | parameter which is proportional to intensity of comparison beam |
| $\lambda$ | (4) | parameter of plate sensitivity |
| $\sigma$ | (2) | scale parameter for exponential profile of a star |
| $\chi$ | (6) | parameter of sensitivity of IDP photomultiplier |

*Table I.*

Definitions for variables used in the text. In parenthesis, before the definition, is the number of the equation in the text which best compliments the definition. Quantities with a "$B$" subscript refer to the background. Primed quantities refer to the value the variable would have if the background transparency equaled some standard background intensity, with no other differences. Greek letters and "const" refer to parameters whose value is usually unimportant, and which do not vary for a given plate. All iris radii refer to the projected radius of the iris' image on the plate. In many cases, the iris radii are reported in some convenient yet arbitrary unit.

$T(r) = 1$). For the case where $R_s > R_{\mathrm{TOE}}$, it is shown in the appendix that $Q$ and $R_s^2$ are linearly related. Since the most common type of IDP is the VIDP, all the results in this paper will assume a VIDP is used.

## III. Calibration Curve

Finding the general form of a calibration curve is the single most useful application of the method outlined in section II. From knowledge of the general form, a fit to the IDP data for the comparison stars becomes more accurate, especially when the data are noisy or there are few comparison stars. A knowledge of the general form of the calibration curve will allow extrapolation to magnitude ranges for which no comparison stars are measured.

This section will show three models (for different regimes of exposure) and their resulting calibration curves. Table II summarizes the assumptions made in calculating the model for each region. Table III summarizes the results on the form of the calibration curve for each regime of exposure. In general, within each regime the calibration curve will be an equation relating $R_s$ to $m$, with three parameters (one parameter for the size scale of the star's profile, one parameter for the zero of the magnitude scale, and one parameter for the sensitivity of the plate).

TABLE II

Summary of Model Assumptions

| Region Number | Assumed Star Profile | Assumed Characteristic Curve | Background Illumination? |
|---|---|---|---|
| Region 1 | Gaussian | Linear | yes |
| Region 2 | Exponential | Linear | no |
| Region 3 | Power Law | Power Law | no |

TABLE III

Summary of Model Results

| Region of Star's Profile | Typical Magnitude Range | Form of Calibration Curve | Parameters | Reference to Equations |
|---|---|---|---|---|
| Gaussian Unsaturated | $m > m(\ell_J)$ | $R_s^2 - R_B^2$ is nearly exponential in m | $R_*, m(\ell_J), R_B$ | (18) |
| Gaussian Saturated | $6 \gtrsim m(\ell_J) - m > 0$ | $R_s^2$ is linear in m | $R_*, m(\ell_J), R_B$ or $R_*$, const | (16),(17) |
| Exponential Profile | $10 \gtrsim m(\ell_J) - m \gtrsim 6$ or $5'' > R_s \gtrsim 2''$ | $R_s^2$ is quadratic in m | $\sigma$, const, const | (19) |
| Power Law Profile | $R_s$ is greater than 5" | $\ell n\ R_s$ is linear in m | $\eta$, $\gamma$, const. or $\eta$, $R_B$, const | (20-22) |

*Region 1.*

For the first model we assume that the incoming starlight's intensity profile is Gaussian (a good assumption for stars within roughly six magnitudes of $m(\ell_J)$) plus a background component,

$$I(r) = I_B + \frac{\ell}{2\pi R_*^2} \exp\left( -\frac{r^2}{2R_*^2} \right) \quad , \tag{11}$$

and that the plate's characteristic curve is well represented by a linear function (eq. (4)). For convenience, the variable $W$ is defined as

$$W = \frac{\lambda\ell}{2\pi R_*^2} \quad . \tag{12}$$

To derive the calibration curve, (1) substitute (11) in equation (4); (2) substitute this result in equation (9); (3) perform the integral; (4) eliminate $\kappa$ with equation (10). For saturated stars (i.e., stars whose images are saturated at the center $[m < m(\ell_J)]$),

$$R_s^2 = R_B^2 + 2R_*^2 \left[ \ell n \frac{W}{T_B} + 1 - \frac{W}{T_B} \exp\left( -\frac{R_s^2}{2R_*^2} \right) \right] \quad , \tag{13}$$

while for unsaturated stars ($m > m(\ell_J)$),

$$R_s^2 = R_B^2 + 2R_*^2 \frac{W}{T_B} \left[ 1 - \exp\left( -\frac{R_s^2}{2R_*^2} \right) \right] . \tag{14}$$

Using the formula for intensity ratios,

$$\ell n\left( \frac{\ell}{\ell_J} \right) = 0.9210[m(\ell_J) - m] \quad , \tag{15}$$

we can find the calibration curve. For saturated stars,

$$R_s^2 = [2R_*^2 + R_B^2 + 1.8420\, R_*^2\, m(\ell_J)] - (1.8420\, R_*^2) m - 2R_*^2\, 100^{[m(\ell_J) - m]/5} \cdot \exp\left( \frac{-R_s^2}{2R_*^2} \right) \tag{16}$$

which reduces to

$$R_s^2 = \text{const} - (1.8420)\, R_*^2\, m \tag{17}$$

as found by Stock (1958). This result is important because it shows (to a first approximation) that a calibration curve using $R_s^2$ is linear in $m$ over a wide range of magnitudes. The calibration curve for unsaturated stars is

$$R_s^2 = R_B^2 + 2R_*^2\, 100^{[m(\ell_J) - m]/5} \cdot \left[ 1 - \exp\left( \frac{-R_s^2}{2R_*^2} \right) \right] \quad . \tag{18}$$

Within this model, all calibration curves may be characterized by three parameters of the system ($R_*$, $R_B$, and $m(\ell_J)$—all of which can be measured separately). For most applications, the calibration curve will be in region 1 (the exceptions occur when bright stars are being measured or the plate has a large plate scale).

*Region 2.*

For stars brighter than those considered in model 1, the profile of the incoming starlight falls off exponentially for intensities where the characteristic curve is in its linear region. The central (Gaussian) portion of the star image is saturated, so there is no significant difference between the real and the model's value for $T(r)$. Using equations (2), (4), (9), (10), and (15), the calibration curve is found to be of the form:

$$R_s{}^2 = \text{const} + \text{const}\ m + \sigma^2\,(0.8482)m^2 \quad . \tag{19}$$

For the case with $R_s < R_{\text{TOE}}$, the same result is valid (with different values for the constants).

*Region 3.*

For highly overexposed stars, the power-law star profile (eq. (3)) is appropriate. Using the power-law characteristic curve (eq. (5)) and equations (9), (10), and (15), the calibration curve is found to be

$$R_s{}^2 = R_B{}^2 + \text{const}\ 100^{-2m/5\eta} \quad . \tag{20}$$

For stars this bright, the approximation $R_s \gg R_B$ is good. This yields

$$\ell n\, R_s = \text{const} - \frac{0.9210}{\eta}\, m \quad , \tag{21}$$

Bright stars have large images, so usually $R_s < R_{\text{TOE}}$. For this case the calibration curve is

$$\ell n\, R_s = \text{const} - \frac{(0.9210)\gamma}{\eta\gamma + 2}\, m \quad . \tag{22}$$

To test the predictions, a plate of the North Polar Sequence has been measured over a wide range of magnitudes. These data are tabulated in Table IV (magnitudes with an asterisk were taken from Selected Area number 1, and show a larger scatter than stars from the North Polar Sequence) and shown in Figure 1. The regime of exposure of this plate is the exponential and power-law segments. The plate DNY 43 (Schaefer 1979, see Table X and Fig. 2) also covers a wide range of magnitudes, in the Gaussian and exponential regimes of exposure. Table V gives the model parameters for the calibration curves of nine plates for which data appear in the literature. The ninth column gives the average deviation from the model calibration curve (note that the plates from Purbosiswojo (1963) and Schaefer (1979) were chosen for the severity of their background variation). For each of these

TABLE IV

Plate MC 114

| $m_{pg}$ | $R_s'$ | $m_{pg}$ | $R_s'$ |
|---|---|---|---|
| 6.46 | 95.0 | 13.22* | 40.1 |
| 6.47* | 99.0 | 13.22* | 42.0 |
| 7.11 | 85.0 | 13.52 | 40.4 |
| 7.38 | 89.1 | 13.59 | 38.7 |
| 8.32 | 79.0 | 13.78* | 38.2 |
| 8.93 | 71.2 | 14.04 | 36.0 |
| 8.96 | 70.4 | 14.10 | 35.1 |
| 9.11 | 69.7 | 14.49* | 33.3 |
| 9.18 | 65.1 | 14.50 | 34.0 |
| 9.77* | 63.9 | 14.61 | 32.2 |
| 9.86* | 65.2 | 14.73* | 32.1 |
| 9.86* | 62.2 | 14.76* | 30.4 |
| 10.32* | 62.3 | 14.80 | 29.6 |
| 10.39* | 60.5 | 15.30 | 28.7 |
| 10.45* | 58.8 | 15.31 | 28.6 |
| 10.77* | 58.8 | 15.35* | 27.8 |
| 10.87 | 55.9 | 15.60 | 27.1 |
| 10.92* | 58.4 | 15.87* | 25.9 |
| 11.05* | 55.9 | 15.94* | 25.7 |
| 11.15* | 58.8 | 15.94* | 25.0 |
| 11.47* | 56.0 | 15.94* | 24.8 |
| 11.64* | 51.3 | 15.94 | 24.3 |
| 11.74* | 52.7 | 16.02 | 26.0 |
| 12.32 | 44.5 | 16.19 | 24.5 |
| 12.68 | 44.7 | 16.40 | 22.7 |
| 12.98* | 42.8 | 17.06 | 22.5 |
| 13.04 | 42.4 | 17.24 | 21.7 |

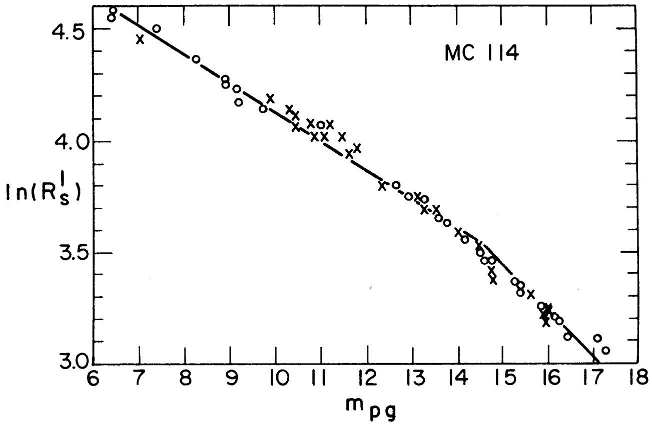


FIG. 1—The calibration curve for the plate MC 114 (see Table IV). $R_s' = 36$ corresponds to an angular size of 5″ and a magnitude of 14$^{m}$5. Stars brighter than this are in the $\eta = 5$ power-law regime, while fainter stars are in the exponential regime.

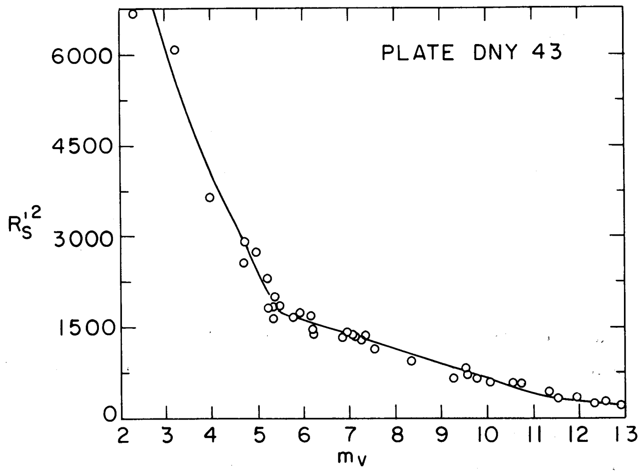


FIG. 2—The calibration curve for the plate DNY 43 (data from Schaefer (1979) is given in Table X). The model calibration curve (shown as a smooth line) is described in Table V. The parameters for the Gaussian region were found by fitting stars with $6.2 < m < 9.8$. This fit was then extrapolated for the remainder of the Gaussian region (three magnitudes). Brighter than 5$^{m}$5, the calibration curve was fitted to equation (19) (exponential region), using the value of $\sigma$ found by King (1971). The brighter stars on this plate (as well as on plate DNY 256) have systematically large field errors because they were chosen from all over the area of the plate, while the faint stars were chosen from one area of the plate.

TABLE V

Calibration Curves for Published Data

| Plate | Source | Region | $R_B'$ [1] | $R_*$ | $m(\ell_j)$ | $\eta$ | $\gamma$ | $\Delta m$ [2] |
|---|---|---|---|---|---|---|---|---|
| Franklin-Adams plates | Kwee (1962) | Gaussian saturated | 3 | 3 | 3 | -- | -- | 4 |
| NGC 6530 | Purbosiswojo (1963) | Gaussian saturated | 8[5] | 3.42 | $12^m.96$ | -- | -- | $0^m.32(\sim1^m)$ |
| U 2858 | Moffat (1969) | Power Law | -- | -- | -- | 5 | 0.96 | 4 |
| V 2835 | Moffat (1969) | Power Law | -- | -- | -- | 5 | 0.96 | 4 |
| PKS0735+178 | Schaefer (1979) | Gaussian unsaturated | 43.8[5] | 33[6] | $14^m.77$ | -- | -- | $0^m.06(0^m.23)$ |
| DNY 256 | Schaefer (1979) | Gaussian | 60[5] | 25.3[6] | $10^m.0$[7] | -- | -- | $0^m.31(\sim1^m)$ |
| PKS0736+017 | Schaefer (1979) | Gaussian unsaturated | 36.8[5] | 23 | $14^m.87$ | -- | -- | $0^m.09(0^m.25)$ |
| DNY 43 | Schaefer (1979) | Gaussian and exponential[8] | 14[5] | 11.3[9] | $11^m.06$[9] | -- | -- | $0^m.16(0^m.42)$ |
| MC 114 | This paper | Exponential[8] and Power Law | 20.5 | -- | -- | 5 | 0.7 | $0^m.16$ |

[1] $R_B'$ is a measured quantity for all plates.

[2] In parenthesis, is the value of $\Delta m$ before the background correction was applied.

[3] Kwee implies that his iris readings are linear in $R_s$. The model for region 1, then implies that this calibration curve must be a parabola (as seen). More detail cannot be given without knowing $R_B$ and $R_*$.

[4] The data fits the model to within the error from which the data may be read off of the published graph.

[5] Data for all stars on the plate has been background corrected to this value.

[6] Value of $R_*$ found from the $R_I$ versus m curve.

[7] Value of $m(\ell_j)$ judged by eye from examination of the star images.

[8] The value of $\sigma$ was chosen to agree with that found by King (1971).

[9] These values were found by fitting the model to stars with $6.2 < m < 9.8$.

plates the scatter around the model calibration curve is similar to the scatter around the "best" drawn calibration curve.

As another application, the known form of the calibration curve can be used to extrapolate the curve where magnitudes are not known. Care must be taken that the extrapolation does not cross a transition point between regimes of a star's profile. An example, based on data taken by Ralph and Johnston (1979), is given in Table VI. While searching for the optical counterpart of the X-ray source A0538-66, they measured a series of stars on two plates. Their sequence of comparison stars extended only as faint as fourteenth magnitude, yet the candidate was still fainter. They also measured faint stars for which no magnitudes were known, but did not measure $R_B$. The form of the calibration curve (eq. (17)) being known, it can be confidently extrapolated. For plates 1 and 2, a calibration curve of $R_s^2 = (0.01021)-(0.0005125)m$ and $R_s^2 = (0.01187)-(0.000610)m$, respectively, yields the magnitudes given in Table IX. These calibration curves were found by requiring that the derived magnitudes for all the stars (except the candidate) be as close as possible for the two plates. The average difference in magnitude for the unknown stars is a tenth of a magnitude. The candidate changed by over a magnitude. Here the range of extrapolation is equal to the range for which the data are available. (Also note that for plate DNY 43, the fit for the Gaussian portion was made over a $3^m.6$ range, and then extrapolated three more magnitudes.)

In general, the first step of using the model's calibra-

TABLE VI

Extrapolation of Calibration Curve

| Magnitude or Designation | $R_s$ (mm) Plate 1 | m Plate 1 | $R_s$ (mm) Plate 2 | m Plate 2 |
|---|---|---|---|---|
| 11.38 | 0.0665 | 11.29 | 0.0695 | 11.54 |
| 11.76 | 0.0655 | 11.55 | 0.0685 | 11.76 |
| 12.26 | 0.0630 | 12.18 | - | - |
| 12.67 | 0.0605 | 12.78 | 0.0645 | 12.64 |
| 13.19 | 0.0585 | 13.24 | 0.0627 | 13.02 |
| 13.60 | 0.0560 | 13.80 | 0.0590 | 13.75 |
| 13.63 | 0.0570 | 13.58 | 0.0590 | 13.75 |
| 14.02 | 0.0550 | 14.02 | 0.0585 | 13.74 |
| 14.21 | 0.0530 | 14.44 | 0.0582 | 13.91 |
| #P1 | 0.0605 | 12.78 | 0.0630 | 12.95 |
| #P4 | 0.0517 | 14.71 | 0.0535 | 14.76 |
| #P2 | 0.0500 | 15.05 | 0.0515 | 15.11 |
| #P3 | 0.0450 | 15.97 | 0.0465 | 15.92 |
| #D | 0.0395 | 16.87 | 0.0385 | 17.03 |
| A0538-66 | 0.0470 | 15.62 | 0.0545 | 14.59 |

tion curves is to find the relation between the IDP reading and the radius of the diaphragm's image at the plate. Next, the regime(s) of exposure which the sequence is in must be determined (see next paragraph). Then, referring to Table III, choose which equation represents the calibration curve for the appropriate region. In most cases, the data should be fit to this equation, treating the model parameters as being free parameters. Often, more information is available, which can be used to eliminate degrees of freedom from the fit. For example, $R_B$ is easy to measure and both $\eta$ and $\sigma$ are known from whichever regime the star is in. The model calibration curve for DNY 256 had all the degrees of freedom eliminated, yet still fits the data as well as the noise (due to extreme background problems) permits.

The regime of exposure for a given star will depend on $R_s$, $R_{\rm TOE}$ and the plate scale. The question is how far out on the star's profile will contribute to the integral in equation (6) or (9). If $R_s < R_{\rm TOE}$ (i.e., some of the visible star image is outside the diaphragm), then the angular size to which $R_s$ corresponds is the quantity which must be compared to the star's profile. For example, with plate MC 114, a tenth magnitude star has $R_s' = 62$ which corresponds to 9.0 arc seconds, which places the star in the $\eta = 5$ power-law region. If $R_s > R_{\rm TOE}$, then the angular size which corresponds to $R_{\rm TOE}$ must be compared to the star's profile. It must be remembered that the boundary between the regimes of a star's profile are not sharp, and the position of the boundary itself may change slightly with atmospheric or telescopic conditions.

## IV. Correcting for Variable Background

The results from model 1 allow for correcting for the effects of variable background on the plate. These variations can occur for many reasons (e.g., nebulosity or scattered light). To be explicit, consider a plate where the background level changes from star to star. For an individual star (with a measured $R_B$ and $R_s$), either equation (13) or (14) will be valid. Now imagine a plate taken such that the background does not vary (but all other conditions are identical). Each star on such an ideal plate will have an $R_s'$ and $R_B'$ which can be related by equations similar to (13) and (14):

$$R_s'^2 = R_B'^2 + 2R_*^2\left[ \ln\left(\frac{W}{T_B'}\right) + 1 - \frac{W}{T_B'}\exp\left(\frac{-R_s'^2}{2R_*^2}\right)\right] \tag{23}$$

if saturated, or

$$R_s'^2 = R_B'^2 + 2R_*^2\frac{W}{T_B'}\left[1 - \exp\left(\frac{-R_s'^2}{2R_*^2}\right)\right] \tag{24}$$

if unsaturated. A relation can be found between $W/T_B'$ and $W/T_B$ by using equation (10):

$$\frac{W}{T_B'} = \frac{W}{T_B}\left(\frac{R_B'}{R_B}\right)^2 \quad . \tag{25}$$

The procedure for correcting for background consists of: (1) numerically solving either equation (13) or (14) for $W/T_B$; (2) finding $W/T_B'$ from equation (25); (3) numerically solving either equation (23) or (24) for $R_s'$. The background corrected calibration curve is then a plot of $R_s'^2$ versus $m$. For this procedure, the choice of $R_B'$ is arbitrary. $R_B'$ should be chosen so as to minimize the size of the correction. If the magnitude of one object is desired, choose $R_B'$ as equal to the $R_B$ value for the object of interest. If the magnitudes of many objects are desired, it is best to set $R_B'$ equal to a value in the middle of the range of $R_B$ values. In the case where $R_s \gg R_*$, the results become particularly simple. For saturated stars,

$$R_s'^2 = R_s^2 - R_B^2 + R_B'^2 + 4R_*^2 \ln\left(\frac{R_B'}{R_B}\right) \tag{26}$$

while for unsaturated stars,

$$R_s' = \frac{R_s R_B'}{R_B} \tag{27}$$

(compare with eqs. (8) and (9) of Schaefer 1979).

Tables VII through XI list data from Purbosiswojo (1963) and Schaefer (1979) for five plates which have severe background variations. The raw data were often so bad that $R_*$ could only be estimated to within a factor of two. In such cases, one must first apply the correction using an estimated $R_*$. The value for $R_*$ can now be determined quite well from the calibration curve and the correction can then be reapplied to the raw data. The uncorrected and corrected calibration curves for four of these plates are shown in Figures 3 and 4 (the model cal-

TABLE VII

PKS 0735+178 Region

| $M_B$ | $R_S$ | $R_B$ | $R_S'$ |
|---|---|---|---|
| 14.37 | 65.83 | 42.45 | 68.38 |
| 14.81 | 57.15 | 42.14 | 59.93 |
| 15.50 | 50.50 | 41.67 | 53.60 |
| 15.64 | 53.43 | 45.6 | 51.00 |
| 15.77 | 51.25 | 44.0 | 50.98 |
| 16.05 | 52.80 | 47.48 | 48.29 |
| 16.62 | 48.65 | 45.83 | 46.35 |
| quasar | 49.77 | 43.8 | 49.77 |

TABLE VIII

Plate DNY 256

| $M_B$ | $R_S$ | $R_B$ | $R_S'$ |
|---|---|---|---|
| 5.30 | 60.7 | 25.2 | 94.6 |
| 5.73 | 96.7 | 61.4 | 95.5 |
| 6.07 | 67.7 | 29.9 | 95.6 |
| 7.1 | 105 | 68.9 | 97.5 |
| 7.3 | 83.9 | 52.7 | 90.6 |
| 7.5 | 86.9 | 64.1 | 82.9 |
| 7.5 | 87.0 | 64.1 | 83.0 |
| 7.9 | 98.1 | 69.6 | 89.3 |
| 8.0 | 80.9 | 57.2 | 83.7 |
| 8.3 | 92.9 | 69.6 | 83.6 |
| 8.5 | 91.3 | 69.6 | 81.8 |
| 8.6 | 80.5 | 59.3 | 81.2 |
| 8.8 | 75.0 | 58.0 | 77.1 |
| 8.9 | 88.4 | 68.9 | 79.4 |
| 9.1 | 86.6 | 69.6 | 76.6 |
| 9.4 | 61.4 | 49.0 | 74.2 |
| 9.4 | 83.0 | 68.3 | 74.0 |
| 9.8 | 61.3 | 49.0 | 74.1 |
| 9.8 | 78.8 | 66.5 | 71.5 |
| 10.0 | 76.3 | 67.6 | 67.3 |
| 10.3 | 77.5 | 68.9 | 67.0 |
| 10.5 | 75.1 | 67.6 | 66.0 |
| 10.8 | 54.0 | 49.0 | 66.1 |
| 11.0 | 73.9 | 67.6 | 65.5 |
| 11.0 | 53.1 | 49.0 | 65.0 |
| 11.0 | 52.8 | 49.0 | 64.6 |
| 11.3 | 52.9 | 49.0 | 64.8 |
| 11.7 | 50.7 | 49.0 | 62.1 |
| 12.0 | 51.3 | 49.0 | 62.9 |

TABLE IX

PKS 0736+017 Region

| $M_B$ | $R_S$ | $R_B$ | $R_S'$ |
|---|---|---|---|
| 15.09 | 48.1 | 36.5 | 48.53 |
| 15.51 | 42.4 | 35.9 | 43.56 |
| 15.54 | 45.0 | 39.3 | 41.93 |
| 16.05 | 41.2 | 37.1 | 40.84 |
| 16.68 | 42.9 | 40.8 | 38.56 |
| 16.70 | 39.1 | 37.1 | 38.77 |
| 16.87 | 36.6 | 35.3 | 38.20 |
| quasar | 38.9 | 36.8 | 38.90 |

TABLE X

Plate DNY 43

| $m_V$ | $R_S$ | $R_B$ | $R_S'$ |
|---|---|---|---|
| 2.32 | 81.5 | 13.3 | 81.8 |
| 3.24 | 78.5 | 15.3 | 78.0 |
| 4.0 | 60.5 | 14.1 | 60.4 |
| 4.76 | 50.6 | 13.9 | 50.7 |
| 4.79 | 55.1 | 16.0 | 53.9 |
| 5.03 | 53.8 | 16.5 | 52.3 |
| 5.24 | 48.0 | 15.4 | 48.1 |
| 5.3 | 42.8 | 14.2 | 42.6 |
| 5.4 | 41.1 | 14.6 | 40.6 |
| 5.4 | 44.2 | 15.4 | 43.2 |
| 5.42 | 46.6 | 16.6 | 44.7 |
| 5.50 | 43.9 | 15.2 | 43.0 |
| 5.82 | 41.1 | 14.4 | 40.8 |
| 5.96 | 42.6 | 15.4 | 41.5 |
| 6.20 | 40.9 | 13.9 | 41.0 |
| 6.23 | 36.6 | 12.1 | 38.3 |
| 6.25 | 36.0 | 12.3 | 37.5 |
| 6.9 | 36.8 | 14.2 | 36.6 |
| 7.0 | 37.4 | 13.7 | 37.7 |
| 7.1 | 36.7 | 13.3 | 37.3 |
| 7.2 | 36.4 | 13.6 | 36.8 |
| 7.3 | 34.9 | 13.0 | 35.8 |
| 7.4 | 36.7 | 14.0 | 36.7 |
| 7.6 | 34.1 | 14.3 | 33.8 |
| 8.4 | 30.3 | 13.8 | 30.5 |
| 9.3 | 25.4 | 13.8 | 25.6 |
| 9.6 | 28.4 | 13.8 | 28.6 |
| 9.6 | 26.4 | 13.8 | 26.6 |
| 9.8 | 25.5 | 13.8 | 25.7 |
| 10.1 | 24.1 | 13.8 | 24.3 |
| 10.5 | 24.1 | 13.8 | 24.3 |
| 10.8 | 23.2 | 13.8 | 23.4 |
| 11.4 | 19.8 | 13.8 | 20.1 |
| 11.6 | 18.1 | 13.8 | 18.4 |
| 12.0 | 17.9 | 13.8 | 18.2 |
| 12.4 | 16.7 | 13.8 | 17.0 |
| 12.6 | 16.7 | 13.8 | 17.0 |
| 13.0 | 15.1 | 13.8 | 15.3 |

ibration curve is plotted with the background corrected data). The scatter about the corrected calibration curves is comparable to the scatter that would be expected if there were no background variations.

## V. Conclusions

This paper has presented a practical method for the solution of various problems encountered while using an IDP. The general shape of the calibration curve was derived in section three. A method for correcting IDP

TABLE XI

Purbosiswojo's Plate

| $M_B$ | $R_S$ | $R_B$ | $R_S'$ |
|---|---|---|---|
| 9.11 | 13.13 | 7.65 | 13.41 |
| 9.24 | 14.40 | 9.98 | 12.71 |
| 9.29 | 13.62 | 8.52 | 13.19 |
| 9.61 | 12.99 | 8.52 | 12.54 |
| 10.23 | 11.61 | 7.36 | 12.19 |
| 10.28 | 13.50 | 9.91 | 11.75 |
| 10.41 | 13.76 | 9.91 | 12.05 |
| 10.52 | 11.60 | 7.95 | 11.65 |
| 10.70 | 11.10 | 7.30 | 11.75 |
| 11.22 | 10.55 | 7.63 | 10.92 |
| 11.49 | 10.77 | 7.99 | 10.78 |
| 11.63 | 9.73 | 6.64 | 11.10 |
| 11.84 | 12.20 | 10.32 | 9.72 |
| 11.97 | 12.10 | 10.48 | 9.38 |
| 12.43 | 9.52 | 7.39 | 10.18 |
| 12.89 | 9.27 | 7.46 | 9.87 |
| 12.89 | 8.86 | 6.94 | 10.0 |

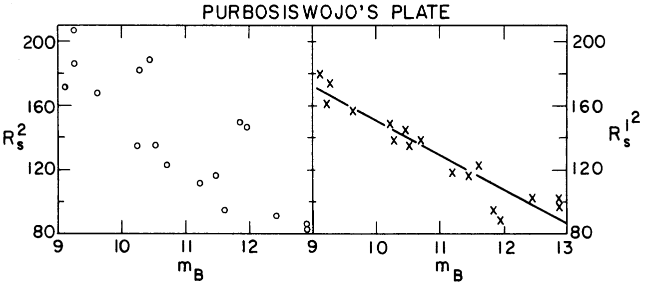


FIG. 3—Data from Purbosiswojo (1963) with and without background corrections. The data are given in Table XI and the parameters of the model calibration curve are given in Table V.

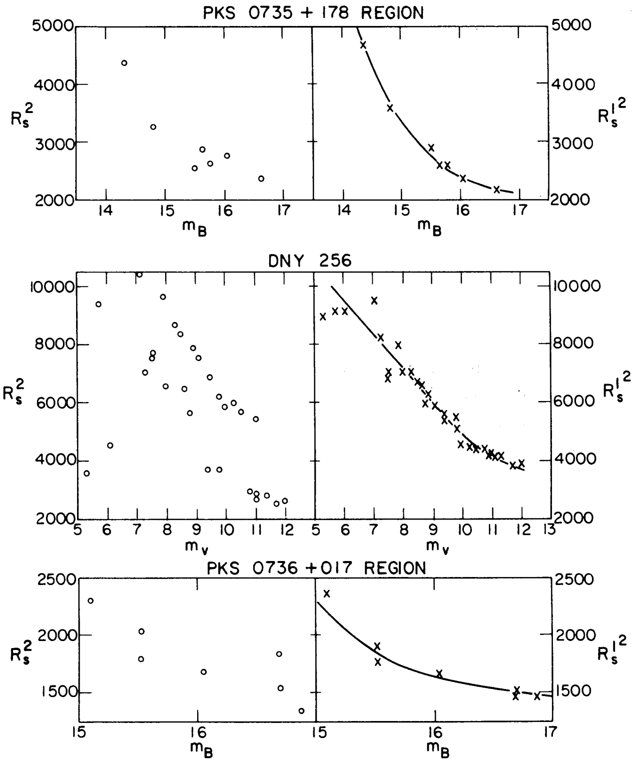


FIG. 4—Data from three plates of Schaefer (1979) with and without background corrections. The data for these three plates are given in Tables VII through IX while the parameters of the model calibration curve are given in Table V. To extend the calibration curve for plate DNY 256, the plate was remeasured, and nine new stars were added (one of Schaefer's (1979) points was deleted due to misidentification of the star).

readings for variable background was demonstrated in section four. Solutions to several more problems (e.g., finding $R_I$ as a function of $m$, or finding $R_0$) are given in the appendix. The predictions of this method have been checked successfully against data taken from Argue (1960), Blaauw (1940), Chun and Freeman (1978), King and Raff (1977), Kwee (1962), La Bonte (1970), Liller and Liller (1975), Moffat (1969), Purbosiswojo (1963), Ralph and Johnston (1979), Schaefer (1979), Weaver (1947), and Zug (1933), as well as against large amounts of data taken by the author.

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## APPENDIX

### Relation Between FIDP and VIDP Readings

For the case where $R_s > R_{\mathrm{TOE}}$ and $R_R > R_{\mathrm{TOE}}$, the readings made by an FIDP and a VIDP are easily related. $R_s$ or $R_R$ can always be made larger than $R_{\mathrm{TOE}}$ by adjusting the magnification or wedge setting of the IDP. $R_{\mathrm{TOE}}$ may be determined to sufficient accuracy by examining the star image under a microscope. Equations (6) through (10) can be used to derive

$$R_s^2 = R_B^2 + R_R^2 (1 - Q) \quad . \tag{28}$$

So the results of measurements made on one type of IDP can be easily converted into what the results would be if the same measurements were made on the other type IDP. Equation (28) qualitatively explains the results of Chun and Freeman (1978), unfortunately quantitative checks are not possible, as they do not quote values of $R_B$ or $R_R$.

### Correcting for a Nearby Star

Occasionally, an observer might want to find the magnitude of a star which has a "nearby star" of known magnitude. A "nearby star" is one whose image can be fit into the iris aperture along with the star of interest, yet the two images do not overlap. The VIDP reading when both stars are in the iris is called $R_2$. Since the magnitude of the "nearby star" is known, the $R_s$ value if this star were measured alone (called $R_c$) can be found from the calibration curve. Equations (9) and (10) can be applied repeatedly to find the $R_s$ value for the star *if* the companion star were not there

$$R_s^2 = R_B^2 + R_2^2 - R_c^2 \quad . \tag{29}$$

Table XII presents data taken for pairs of stars which were far enough apart for each individual image to be measured separately. The results show the formula to be accurate to within measurement errors.

### Correcting for Changing Comparison Beam Intensity

At times, it is necessary to compare readings of stars on a plate taken at different times or with different IDP's. In such cases, it is likely that the intensity of the comparison beam ($\kappa$) will change (Kwee 1962; Argue 1960). Assuming $R_s > R_{\mathrm{TOE}}$, equations (9) and (10) may be applied to the two cases of different $\kappa$ to find a relation between the measured $R_s$ values;

$$(R_s^2 - R_B^2)\Big|_{\kappa_1} = (R_s^2 - R_B^2)\Big|_{\kappa_2} \quad . \tag{30}$$

So if the calibration curve is plotted as $R_2^2$ versus $m$, then a change of $\kappa$ will shift the calibration curve parallel to the $R_s^2$ axis. This upward shift is illustrated in Figure 5.

TABLE XII

Correcting for Nearby Stars

| $R_B$ | $R_c$ | $R_2$ | $R_s$ | $\sqrt{R_B^2+R_2^2-R_c^2}$ |
|---|---|---|---|---|
| 605±15 | 662±15 | 750±15 | 690±15 | 700±26 |
| 560 | 690 | 735 | 590 | 614 |
| 615 | 700 | 750 | 670 | 671 |
| 109±3 | 111±3 | 120±3 | 116±3 | 118±5 |
| 109 | 116 | 118 | 113 | 111 |
| 107 | 114 | 126 | 125 | 120 |
| 105 | 116 | 132 | 127 | 122 |
| 105 | 119 | 132 | 119 | 119 |
| 104 | 110 | 121 | 116 | 116 |
| 106 | 120 | 121 | 114 | 115 |
| 106 | 133 | 139 | 120 | 113 |

### Radius of a Star Image

It is often easy and convenient to measure the radius of a star image ($R_I$) instead of using an IDP. To be more precise let $R_I$ be defined as the distance from the center of the image to where $T$ has risen to some predetermined value,

$$T(R_I) = \text{const} \quad . \tag{31}$$

For any choice of characteristic curve, this can be changed into:

$$I(R_I, \ell) = \text{const} \quad . \tag{32}$$

Take any of equations (1), (2), or (3), set $r = R_I$, use equations (15) and (32), then solve for $R_I$. In each of the three regimes of a star's profile, a relation between $R_I$ and $m$ can be found:

Gaussian:

$$R_I^2 = \text{const} - (1.8420) R_*^2 m \tag{33}$$

Exponential:

$$R_I = \text{const} - (0.9210)\sigma m \tag{34}$$

Power Law:

$$\ell n\, R_I = \text{const} - (0.9210/\eta) m \quad . \tag{35}$$

For the Palomar Sky Survey, King and Raff (1977), Liller and Liller (1975), La Bonte (1970), and others have measured $R_I$ as a function of magnitude. Their results can be summarized as:

Magnitude 14 to 20:

exponential growth ($\sigma = 10.9\mu$)

Magnitude 8 to 14:

power law ($\eta = 5$)

Magnitude −1 to 8:

power law ($\eta = 2$) .

This behavior is readily derived from King's (1971) profile of a star:

1″ to 5″:

exponential growth ($\sigma = 11.1\mu$)

5″ to 15″:

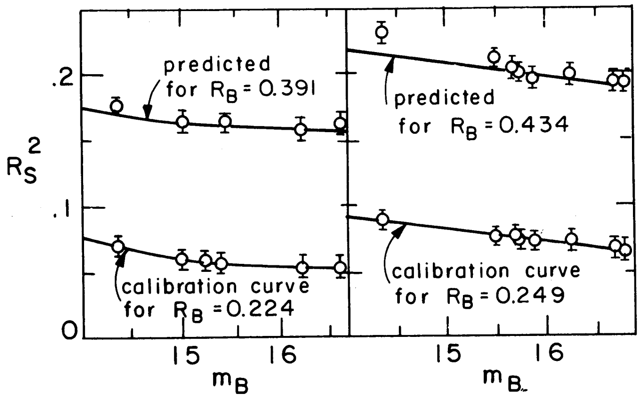


FIG. 5—Two measured calibration curves for two different plates where the parameter $\kappa$ was changed (by adjusting the intensity of the IDP's comparison beam). Equation (30) predicts that the upper calibration curve will be identical with the lower curve, except shifted by an amount $R_{B_2}{}^2 - B_{B_1}{}^2$. For the left-hand side, this upward shift is predicted to be 0.103. For the right-hand side, the upward shift is predicted to be 0.126.

power law ($\eta = 5$)

15″ to 300′:

power law ($\eta = 2$) .

The range in magnitudes of each regime is the same measured by the star profile as it is from the $R_I$ versus $m$ curve.

The $R_I$ versus $m$ curve given by Zug (1933) is well explained within seven magnitudes of the plate limit by $R_* = 23.2$ microns.

$R_{1/2}$ for Unsaturated Stars

The half width at half maximum of unsaturated stars ($R_{1/2}$) can be used to estimate $R_*$. The definition of $R_{1/2}$ demands that

$$T(R_{1/2}) = [T(0) + 1]/2 \quad . \tag{36}$$

If we assume a linear characteristic curve (eq. (4)), and a Gaussian profile (eq. (1)), $R_{1/2}$ is found to be

$$R_{1/2} = R_* \sqrt{2\ell n 2} = (1.15)R_* \quad . \tag{37}$$

If a power-law characteristic curve (eq. (5)) is assumed, then $R_{1/2}$ is still approximately equal to $R_*$, but the result is a weak function of $m$, $\gamma$, and $T_B$.

Correcting for Trailing

If a star image is trailed, it will give a different IDP reading than if the image is not trailed. A need for correcting the iris reading of a trailed star image ($R_T$) can arise in the comparison between two plates, or in finding the brightness of a trailed asteroid image (e.g., Pascu 1973). Let us assume the brightness of the starlight hitting the plate is distributed as a two dimensional Gaussian,

$$I(x,y,\ell,a) = \frac{\ell a}{2\pi R^2} \exp\left(\frac{-x^2}{2R^2}\right) \exp\left(\frac{-y^2a^2}{2R_*^2}\right) \tag{38}$$

where $a \leq 1$. Using the power-law model for the characteristic curve (eq. (5)) and equations (9), (10), and (38) we find an equation for $R_T$. If this same star were photographed under identical conditions, except that the image was not trailed, $R_s$ would satisfy the same equation as $R_T$ (with $a = 1$). Combining these two equations, and assuming the star is saturated we obtain

$$R_s^2 = R_T^2 a + R_B^2(1 - a) - 2R_*^2 \ell n\, a \quad . \tag{39}$$

Point of Optimum Density

Weaver (1947) has developed an improved method for photographic transfer of magnitude sequences which he calls the "point of optimum density" method. This method relies on the fact that the iris reading of a star image of a certain brightness will not change, despite small changes in seeing, focusing, or guiding. Small changes in seeing, focusing, and guiding will affect $I(r,\ell)$ only by changing $R_*$. Mathematically this condition can be expressed as:

$$\left.\frac{dR_s}{dR_*}\right|_{R_0} = 0 \quad . \tag{40}$$

The reason for this effect is easy to see. For a bright star under poor seeing conditions (large $R_*$) the region of saturation will be larger, making the star's image larger, and therefore making the measured $R_s$ larger. For a faint star under poor seeing conditions, the small amount of light will be spread out over a larger area with the result being that fewer grains are developable (so the measured $R_s$ will be smaller). For a star of some intermediate brightness, these two effects will cancel. At this "point of optimum density," $R_s$ will be independent of slight changes in $R_*$. For region 1, if we assume $R_s > R_{TOE}$ and equation (40), $R_0$ is found to be

$$R_0^2 = R_B^2 + 2R_*^2 \quad . \tag{41}$$

If $R_s < R_{TOE}$, the assumption of a power-law characteristic curve gives simple results:

$$R_0^2 = \frac{\gamma + 1}{\gamma} 2R_*^2 \quad . \tag{42}$$

Several authors have reported enough results to be able to check equations (41) and (42). Blaauw (1940) presents measured $Q_0$'s which range between 0.808 and 0.875. His Figure 1 shows typical calibration curves, from which the slope will yield $R_*/R_R$. The value of $R_*/R_R$ varies between 0.32 and 0.27. Using equations (28) and (41), we find that these correspond to a predicted range for $Q_0$ of from 0.80 to 0.855. Weaver (1947) has measured $Q_0 = 0.822 \pm 0.009$ for a set of plates from the Mount Wilson 60-inch telescope. The slope of the calibration curve (in Weaver's Fig. 2) shows that $R_*/R_R = 0.300 \pm 0.010$. This corresponds to a $Q_0 = 0.820 \pm 0.14$. Argue (1960) has measured four separate $R_0$'s for different $R_B$ values. The slope of the calibration curves yields $R_* = 32.9$. The $R_B = 84$ calibration curve has its $R_0$ value as $92 \pm 5$ (eg. (41) predicts $R_0 = 96$). The $R_B = 50$ calibration curve has its $R_0$ value at $71 \pm 3$ (eq. (41)) predicts $R_0 = 68.3$). From the text, it appears that $R_s < R_{TOE}$ for the calibration curves with $R_B$ equal to 35 or 25. With $\gamma = 1$ (the value Argue uses on page 193), equation (42) predicts $R_0 = 65.8$. This is within the error for $R_0$ equaling $68 \pm 4$ and $67 \pm 4$ for $R_B$ equaling 35 and 25, respectively.